Dose-agnostic neural deconvolution for low-dose fluorescence microscopy

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Abstract

Fluorescence microscopy is photon-limited: the light that improves signal-to-noise also drives photobleaching and phototoxicity. Computational restoration can recover structure from fewer photons, but the regularization that works at one noise level fails at another, and most learned models are trained or conditioned for a narrow dose range. In practice the photon count is often unknown, varies across the field, and changes with acquisition settings, so per-dose retuning is difficult to deploy. Here we ask whether a single model can cover a wide dose range without dose metadata, and how much photon budget that saves against a well-tuned classical method.

We train one U-Net on mixed-dose patches with no photon-count input and compare it with TV-ADMM deconvolution tuned at every dose across a 100-fold range. To make the comparison operational, we select the dose on validation fields for a target success rate and apply it unchanged to held-out fields. The same model improves over tuned TV-ADMM by 0.93–1.45 dB across the range, needs a 3–3.3× lower photon budget at matched 90% success on several criteria, and, on real noisy/clean pairs (a denoising check), improves over affine-calibrated BM3D by 4.18 dB. Controls show that this gain comes from learned sample structure rather than capacity, and a PSF-mixed variant retains it under an Airy PSF. The dose sweep is simulated and the real gain is in-source; together they show that one model can replace per-dose tuning and that restoration methods should be compared at matched success rather than at a single dose.

Keywords: fluorescence microscopy, deconvolution, low-dose imaging, deep

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learning, uncertainty

1. Introduction

Fluorescence microscopy is limited by the number of detected photons. Raising excitation power or exposure improves signal-to-noise ratio, but it also accelerates photobleaching and phototoxicity, which is especially restrictive for live samples [1]. Computational restoration offers a way out: recover more of the underlying structure from the photons already collected. Deconvolution removes the diffraction-limited point-spread function (PSF) while suppressing shot noise, but the inverse problem becomes increasingly ill-posed as the photon count falls. A regularization strength that preserves fine structure at one dose amplifies noise at a lower dose; one that suppresses noise at low dose erases structure at a higher dose.

Classical deconvolution makes this trade-off explicit. Richardson–Lucy [2, 3], Wiener filtering [4], total variation [5, 6], non-local means [7], and blind deconvolution [8] all depend on a noise model and a regularization weight that must be chosen for a particular signal level; see Ref. [9] for a review. Deep learning replaces the hand-designed prior with one learned from paired images. CARE [10] and its deployment tools [11] have made U-Net restoration routine, and supervised restoration has been extended to cross-modality super-resolution [12]. Noise-level conditioning is standard in image denoising [13], and blind denoisers can estimate it or learn without an explicit noise level [14, 15, 16], but these methods denoise rather than deconvolve and have not been evaluated at a matched success rate across a wide dose range. Model-based deep deconvolution [17], zero-shot deconvolution from noisy images alone [18], uncertainty-guided adaptive scanning [19], and near-zero-photon imaging [20] address complementary parts of the problem. In microscopy, however, the photon count per pixel is often not known precisely, varies across the field, and changes with acquisition settings. Retuning or retraining for every dose is practical only within a narrow range. The practical question for low-dose imaging is whether one model can cover a wide dose range without being told the dose, and how much photon budget that saves against a well-tuned classical method. The contribution is not a new network architecture; it is a controlled test of that question, with the saving measured at matched success rather than at a single dose.

We address this with a controlled study on public microscopy images. One U-Net is trained on patches whose dose is sampled over a 100-fold range; the network receives only the noisy image and has no dose channel, so it must infer the

local noise level from the data. We compare against total-variation (TV) deconvolution with the known PSF, TV denoising, Wiener filtering, and Richardson–Lucy, all tuned per dose; NLM denoising is included in the qualitative and extended comparisons. Splits are by field of view (FOV), and test FOVs are never used for training, tuning, or early stopping. To make the comparison operational, we define a protocol criterion: choose the smallest dose that reaches 90% success on validation FOVs, then apply it unchanged to held-out FOVs. This avoids selecting the operating point on the test data and turns the dose-response curves into a photon-budget comparison.

Across this range, the blind model improves over tuned TV-ADMM by 0.93–1.45 dB at every dose and needs a 3–3.3× lower photon budget at matched 90% success on several criteria. It also transfers to three held-out stains with smaller positive gains. The real-data check tests denoising: with retraining on the same acquisition protocol, the network improves over affine-calibrated BM3D by 4.18 dB, while a budget-matched DnCNN is close behind. Controls separate the source of the gain: a budget-matched phase-scrambled network falls below TV, whereas dose conditioning and per-dose specialists add little, so the gain is not explained by capacity or metadata. A PSF-mixed variant keeps the gain under a physical Airy PSF, where the Gaussian-trained model loses its high-dose advantage (SI S4d).

Practically, this removes one calibration axis. In live imaging the acceptable dose is set by the sample rather than by the restoration algorithm; a model that covers the full range without metadata can be trained once and reused as the protocol changes, and the protocol curve selects the dose that reaches a desired quality. It also changes how restoration methods should be compared: a gain measured at one dose can shrink or grow at another, so the meaningful quantity is the photon budget required at a matched success rate. The scope is limited in two ways: the dose sweep is simulated with a known Gaussian PSF and Poisson noise, and the real pairs test restoration within one acquisition protocol; the same weights do not transfer zero-shot to a different source. Within that scope, the results support dose-agnostic restoration.

2. Methods

Forward model.. A clean FOV x is blurred by a Gaussian PSF of width σ and corrupted by Poisson shot noise,

$$y = \frac{1}{P}\,\mathrm{Poisson}(P\,(g_\sigma * x))\,, \tag{1}$$

where P is a full-scale Poisson scale: a pixel with normalized intensity x has expected detected photons Px. Unless stated otherwise $\sigma = 2$ pixels and P ranges from 3 to 300 in full-scale units. The same PSF is used to generate the data and by all classical baselines. For brevity we call P the dose; the mean detected photons per pixel on the test patches are 0.36, 3.56, and 35.6 at $P = 3, 30, 300$, respectively (Poisson means in `results/photon_units.json`). Because the expected count scales linearly with P, a ratio in P is also a ratio in photon budget.

Data.. We use four public datasets from the Broad Bioimage Benchmark Collection [21]. The main dataset is BBBC014 channel 1 (96 FOVs, 1360×1024 pixels, 8-bit, CC BY 3.0), split at the FOV level into 56 training, 20 validation, and 20 test FOVs. Validation FOVs are used for early stopping and for tuning every classical baseline; test FOVs are never used for training, tuning, or model selection. BBBC015 and BBBC016 (U2OS cytoplasm-to-nucleus assays; CC BY 3.0) and BBBC018 (actin channel; a different cell type and stain; CC BY-NC-SA 3.0) are used only as external test sets, with 20 FOVs sampled uniformly from each (five for TV tuning and 15 held out for evaluation). All images are normalized by their 8-bit full scale.

Real noisy/clean data.. For a real-noise check we use the public AI4Life-MDC25-Nuclei release [22] (Zenodo 15624741, CC BY 4.0), the training set of the 2025 AI4Life Microscopy Supervised Denoising Challenge. It combines 2,457 256×256 pairs from the Jacquemet noisy-nuclei source with 20 1024×1024 pairs from BBBC006; the two sources differ in bit depth, saturation, and noise model. We exclude the 20 small-pair IDs that coincide numerically with the large-field IDs as a precaution (the images are not crops of one another), leaving 2,437 pairs split by image ID into 1,951 training, 243 validation, and 243 test pairs. No field is shared: over all 1,951 training images, the maximum test-train thumbnail correlation is 0.55, no pair exceeds 0.8, and the top pair has full-resolution correlation 0.06 (`results/real_leakage_full.json`). The real-data comparison uses all 243 held-out pairs; classical outputs are affine-calibrated on the validation pairs (Methods, Real-data model). The 20 large BBBC006 pairs are never used for training or tuning and form a cross-source transfer check.

Second real source.. The FMD release [23, 24] (Zenodo 15761900, CC BY 4.0) contains 63,669 256×256 noisy/clean pairs (multiple frames per field) from three microscopes, three channels, and four noise levels. We use one frame from each of 1,998 fields (seed 0) and split by field, stratified by microscope/channel/level, into 1,396/301/301 training/validation/test; no frame of a field crosses splits. The main

comparison uses a pre-specified 150-field subset: the first four fields per stratum in field-ID order, plus six fifth fields, selected before prediction.

Network and training.. The network is a three-level U-Net [25] with 64, 128, and 256 channels and 3×3 convolutions. Training patches are 64×64 pixels drawn from training FOVs, with random flips and 90° rotations. The main mixed-dose model samples the dose of every patch from $\{3, 10, 30, 100, 300\}$ full-scale units and receives only the noisy image; no dose channel is used at training or inference. It uses 48,000 patches and a 150-epoch schedule with early stopping on validation PSNR averaged over all doses. A PSF-robust variant uses the same schedule but randomly blurs each training patch with the Gaussian ($\sigma = 2$) or a scalar Airy PSF (NA 1.4, $\lambda = 520\,\mathrm{nm}$, 100 nm pixels). Optimization uses Adam with a learning rate of 10^{-3}, a cosine schedule, and a mean-squared-error loss. A base-budget ablation uses a 32-channel conditioned variant (24,000 patches, 100 epochs); a budget-matched conditioned model repeats the full 64-channel/48,000-patch/150-epoch schedule. The conditioned variant adds a constant dose channel $c(P) = (\log_{10} P - \log_{10} 3)/(\log_{10} 300 - \log_{10} 3)$. Per-dose specialists use the same 64-channel/48,000-patch/150-epoch schedule, trained at 3, 30, or 100 full-scale units. Two phase-scrambled controls replace every training FOV by a random-phase surrogate with the same power spectrum while keeping validation and test FOVs unchanged: a base-budget control (32 channels, 24,000 patches, 100 epochs) and a budget-matched blind control (64 channels, 48,000 patches, 150 epochs, mixed doses). Validation uses four spread 128×128 patches per FOV.

Real-data model.. The real-data network is a CARE-style supervised U-Net (the same three-level encoder–decoder used for the simulation model, with 48 channels in the first level) and is trained on 128×128 patches from the real training pairs (16,000 patches, 80 epochs, batch 64, Adam with learning rate 10^{-3}, cosine schedule), with flips and 90° rotations. Each image is normalized independently to its 0.5–99.9 percentile range. Validation on the 243 held-out small pairs selects the checkpoint (28.6 dB). For the nuclei pairs, classical baselines are tuned on five validation pairs over extended grids (BM3D [26] $\sigma \in [0.2, 1.0]$, NLM $h \in [0.1, 0.3]$, TV weight $\in [0.08, 0.4]$); the selected values are $\sigma = 0.3$, $h = 0.2$, and $w = 0.24$ (grid means in `results/real_small_tuning_grid.json`). Each classical output is additionally mapped by one global affine transform (gain and offset) fitted on the validation pairs, because the supervised networks are trained on paired images; raw uncalibrated values are in the SI. A second supervised baseline is a 17-layer DnCNN [27] with 64 channels and residual noise prediction, trained

on 16,000 patches for 80 epochs with the same normalization and augmentation (validation 28.2 dB). The FMD model uses the smaller base budget (32 channels, 8,000 patches, 40 epochs) and is therefore a secondary check. Its classical baselines are tuned separately for each microscope on five validation fields: the selected BM3D sigmas are 0.12 (CF), 0.15 (TP), and 8.0 (WF), where the WF search was extended to 20 to avoid a boundary solution; NLM uses $h = 0.1$, and TV weights are 0.08 (CF) and 0.12 (TP, WF). The FMD checkpoint is selected on 60 FMD validation fields. Test pairs are never used for tuning or early stopping.

Baselines.. The primary classical baseline is TV-regularized deconvolution: we solve $\min_x \frac{1}{2}\|Hx - y\|^2 + \lambda\,\mathrm{TV}(x)$ with ADMM [28], the known PSF H, and 30 iterations. Both the TV weight λ and the ADMM penalty ρ are tuned on five validation FOVs (50 patches) at each dose, with reflect padding at the FOV boundary. We also report TV denoising without deblurring (Chambolle TV on y), Wiener filtering, and Richardson–Lucy with the same PSF. NLM after an Anscombe variance-stabilizing transform [29, 30], Wiener, and Richardson–Lucy are included in the Supplementary Information with extended parameter grids, because these baselines either denoise without deblurring or are unstable at very low counts. All classical parameters are tuned on validation FOVs at the same dose and PSF as the network. BM3D is a denoiser without a PSF model, so it is used in the real noisy/clean comparison but not in the simulated deconvolution sweep, where PSF-aware baselines are the appropriate classical comparators.

Metrics and statistics.. We report PSNR, SSIM [31], and a foreground PSNR computed on the top 10% of ground-truth intensities. The main comparison and the protocol use ten spread 128×128 patches per FOV, aggregated per FOV. The protocol dose is the smallest dose in $\{3, 10, 30, 100, 300\}$ full-scale units at which at least 90% of validation FOVs reach a target; it is then applied unchanged to the test FOVs. Test-oracle doses are reported only as an upper bound in the SI. The 90% criterion corresponds to 18/20 FOVs and the five-point grid makes the reported factors discrete. Comparisons use per-FOV paired differences, bootstrap 95% confidence intervals over FOVs, and exact Wilcoxon signed-rank tests; $n = 20$ FOVs. An effect is reported only when the FOV-level test and the confidence interval agree.

Uncertainty.. We train three dose-conditioned networks with different random seeds, following the deep-ensemble construction [32]. At a given dose, the per-pixel standard deviation across the three predictions is used as an uncertainty map; its FOV mean is the uncertainty score. The uncertainty analysis uses the

conditioned variant as a proxy for the mixed-dose model. We evaluate failure detection with the area under the receiver-operating-characteristic curve (AUC) for FOVs whose PSNR is below 34 dB, and rank correlation with reconstruction RMSE. At $P = 30$ only one FOV fell below the target, so its AUC is not reported.

Timing.. Inference time is measured per 128×128 patch on an NVIDIA A100 GPU for the network and on one CPU core for TV deconvolution with 30 ADMM iterations. The network is also timed on one CPU core for a same-hardware comparison.

3. Results

3.1. Fair deconvolution comparison

Table 1 reports all five doses and Fig. 1 shows reconstruction at $P = 3, 10, 30$. At $P = 3$ the mixed-dose network reaches 32.27 dB PSNR and 0.907 SSIM, compared with 31.34 dB and 0.885 for tuned TV deconvolution, 30.31 dB and 0.835 for TV denoising, and 14.60 dB for the noisy input. The foreground PSNR is 25.25 dB versus 24.61 dB for TV deconvolution. The advantage holds at every dose: at $P = 30$ the network reaches 36.70 dB versus 35.34 dB, and at $P = 300$ 40.01 dB versus 38.63 dB. The paired difference over FOVs is 0.93 dB (95% CI 0.83–1.05), 1.12 dB (CI 1.06–1.19), 1.36 dB (CI 1.23–1.49), 1.45 dB (CI 1.34–1.57), and 1.37 dB (CI 1.25–1.51) at $P = 3, 10, 30, 100, 300$; the network wins on all 20 test FOVs at each dose (Wilcoxon $p < 10^{-5}$). SSIM improves from 0.885 to 0.907 at $P = 3$ and from 0.963 to 0.971 at $P = 300$.

Table 1: Reconstruction on 20 held-out BBBC014 FOVs (ten 128×128 patches per FOV). PSNR (dB), SSIM, and foreground PSNR (dB). The U-Net is the mixed-dose model with no dose input. TV deconv is ADMM with the known PSF; TV denoise is Chambolle TV without deblurring. Classical parameters are tuned on validation FOVs at each dose. Paired gains quoted in the text are means of unrounded per-FOV differences and can differ by 0.01 dB from differences of the rounded table entries.

	Noisy	TV denoise	TV deconv	U-Net
$P = 3$ *(full-scale units)*				
PSNR (dB)	14.60	30.31	31.34	32.27
SSIM	0.054	0.835	0.885	0.907
Foreground PSNR (dB)	10.73	23.80	24.61	25.25
$P = 10$ *(full-scale units)*				
PSNR (dB)	19.53	32.50	33.62	34.74
SSIM	0.151	0.905	0.917	0.934
Foreground PSNR (dB)	15.25	25.33	26.77	27.68
$P = 30$ *(full-scale units)*				
PSNR (dB)	24.10	34.08	35.34	36.70
SSIM	0.331	0.923	0.937	0.950
Foreground PSNR (dB)	19.60	26.91	28.39	29.67
$P = 100$ *(full-scale units)*				
PSNR (dB)	28.81	35.36	37.16	38.61
SSIM	0.596	0.935	0.952	0.963
Foreground PSNR (dB)	23.92	28.22	30.22	31.63
$P = 300$ *(full-scale units)*				
PSNR (dB)	32.48	36.26	38.63	40.01
SSIM	0.793	0.947	0.963	0.971
Foreground PSNR (dB)	26.95	29.01	31.67	33.03

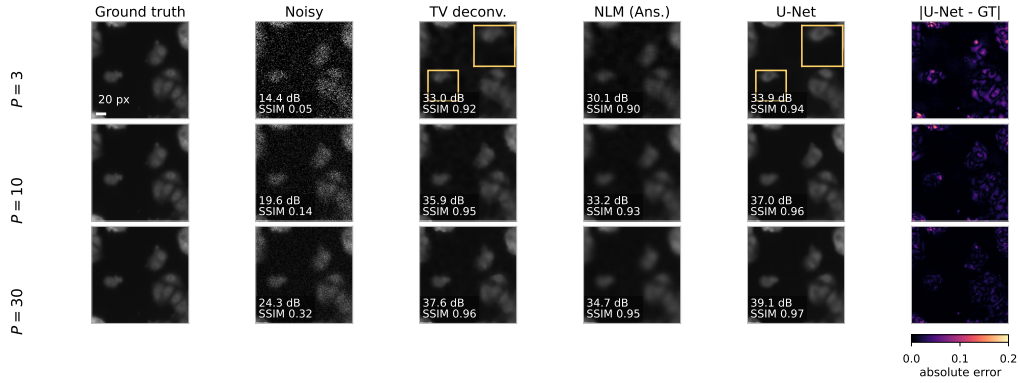


Figure 1: Qualitative reconstruction at $P = 3, 10, 30$ on a fixed 192×192 test crop (seed 7, not selected by performance). Columns: ground truth, noisy input, TV deconvolution, NLM, mixed-dose U-Net, and the absolute U-Net error. Scale bars are 20 pixels; the public BBBC014 release does not include a pixel-size calibration, so physical units are not claimed. The yellow boxes and insets magnify the same 60×60 region of the TV and U-Net panels at $P = 3$ (approximately 1.3×). Panel PSNR values refer to this single crop, not to the FOV means in Table 1; all panels use the same normalized intensity scale $[0, 1]$. The network recovers boundaries and internal texture that TV deconvolution smooths, and the error is concentrated on edges.

TV deconvolution is the strongest classical method at every dose; TV denoising without deblurring is 1.96 dB below the network at $P = 3$, whereas fair TV deconvolution is 0.93 dB below. The gain is 0.93, 1.36, and 1.37 dB at 3, 30, and 300 against TV deconvolution, with no monotonic dose trend; it is largest at $P = 100$ (1.45 dB) and smallest at $P = 3$ (0.93 dB).

3.2. Dose to matched quality

Figure 2a–c shows PSNR, SSIM, and foreground PSNR as a function of dose. Fig. 2e converts these curves into a fixed protocol: choose the smallest dose that reaches 90% success on the 20 validation FOVs, then apply it unchanged to the test FOVs. The network needs a 3× lower dose P (equivalently, a 3× smaller photon budget) at PSNR 32 dB and SSIM 0.90, and 3.3× lower at PSNR 34 dB and foreground PSNR 26 dB; SSIM 0.85, SSIM 0.92, and foreground PSNR 24 dB are ties. At PSNR 30 dB the validation-selected U-Net dose reaches 80% on test, and at foreground PSNR 22 dB the TV dose reaches 80%, so neither factor is claimed (SI S2b). Selecting the dose on the test FOVs gives an upper bound of 3–3.3× at five targets and four ties (SI S2b). The factors are discrete and the

20-FOV sample is small; per-target CIs from the test-oracle bootstrap are reported in SI S2b, where 96.6% of resamples give at least 3× at PSNR 32 dB. The model is also not restricted to the five training doses: at $P = 5, 50, 200$, none of which was used in training, it reaches 33.32, 37.50, and 39.47 dB, within 0.02 dB of log-dose interpolation between the neighboring trained doses (SI S4c).

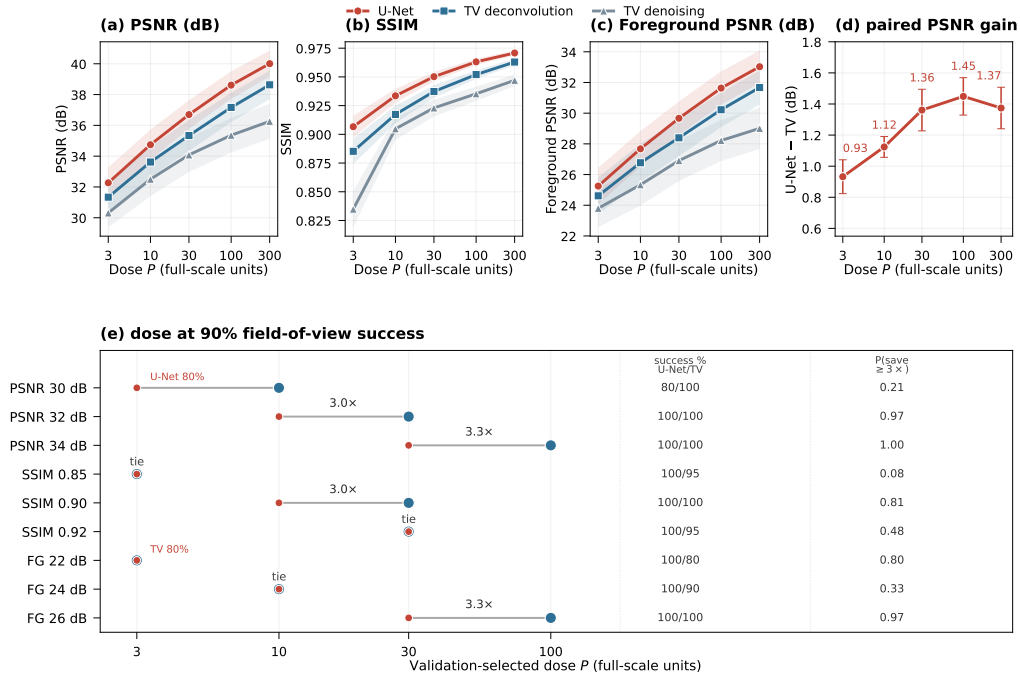


Figure 2: (a–c) PSNR, SSIM, and foreground PSNR versus dose P on 20 held-out FOVs; shaded bands are 95% confidence intervals over FOVs. (d) Paired PSNR gain of the U-Net over TV deconvolution on the same FOVs; error bars are 95% confidence intervals. (e) Validation-selected dose for 90% success on validation FOVs; blue dots are TV deconvolution and red dots the U-Net. Factors are shown only when both methods also meet the 90% criterion on held-out FOVs; "tie" marks equal doses, and labels mark the method whose validation-selected dose reaches only 80% on test. Right-hand text columns give the held-out success rate (U-Net/TV, %) and the FOV-bootstrap probability that the saving is at least 3×. Test-oracle doses and FOV-bootstrap CIs are in SI S2b.

We also compared the mixed-dose model with a dose-conditioned variant and with per-dose specialists. The budget-matched conditioned variant (64 channels, 48k patches, 150 epochs) reaches 32.29, 36.71, and 38.66 dB at $P = 3, 30, 100,$

compared with 32.27, 36.70, and 38.61 dB for the blind model; it is within 0.05 dB (+0.01–+0.05 dB) but requires the photon count at inference. Per-dose specialists trained with the same 64-channel/48k/150 budget reach 32.36, 36.72, and 38.68 dB at their own doses, within 0.1 dB of the blind model, but require one model per dose and the dose label. The blind model is therefore preferred: it is within 0.05 dB of the conditioned variant and within 0.1 dB of the specialists, with no acquisition metadata.

3.3. Transfer across cell types and stains

The model trained on BBBC014 channel 1 transfers without retraining to three held-out datasets with different morphology and staining (Fig. 3a, Table 2). At $P = 30$ it gains 0.52 dB over tuned TV deconvolution on BBBC015, 1.30 dB on BBBC016, and 0.16 dB on BBBC018; the in-domain gain is 1.36 dB. The external confidence intervals exclude zero, but the gains are smaller than in-domain and the BBBC018 gain is modest; transfer helps but does not replace in-domain training.

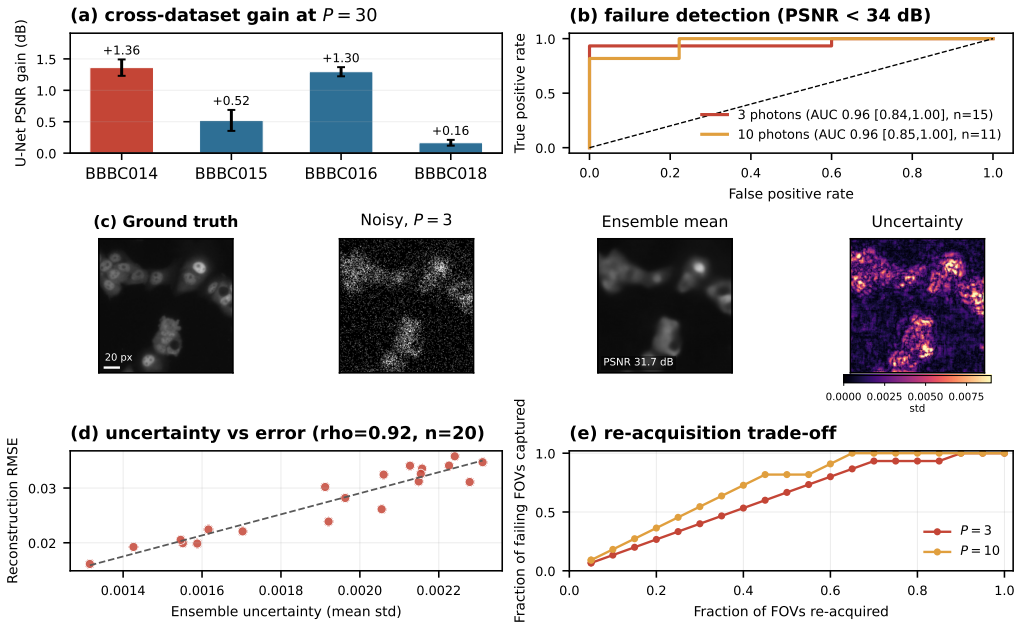


Figure 3: (a) Mean PSNR gain of the mixed-dose network over tuned TV deconvolution at $P = 30$ on four datasets; red is the in-domain dataset and blue the external datasets; whiskers are 95% bootstrap CIs. The external TV weights are tuned on five separate fields per dataset. (b) Failure-detection ROC from a base-budget uncertainty ensemble. (c) Example at $P = 3$ on a validation FOV: ground truth, noisy input, ensemble mean, and uncertainty (standard deviation). (d) Uncertainty versus RMSE on held-out FOVs, with a linear fit. (e) Re-acquisition trade-off: fraction of failing FOVs captured versus fraction of FOVs re-acquired, at $P = 3$ and 10. AUCs and 95% bootstrap CIs are in (b); at 30 only one FOV fails, so no AUC is shown.

3.4. Uncertainty identifies unreliable fields

The three-model ensemble provides a per-pixel uncertainty map (Fig. 3c). At $P = 3$ the uncertainty is concentrated on cell boundaries and on dim structures, where the error is largest. Across FOVs, uncertainty correlates with reconstruction RMSE (Spearman $\rho = 0.92$), and it separates FOVs that fail a 34 dB target from those that succeed with AUC 0.96 (95% CI 0.84–1.00) at $P = 3$ and 0.96 (0.85–1.00) at 10 (Fig. 3b,e; SI S4). At 30 only 1 of 20 FOVs fell below the target, so no AUC is reported. This provides a practical quality-control signal: fields with high uncertainty can be re-acquired or interpreted conservatively without a ground-truth image.

Table 2: Cross-dataset PSNR at $P = 30$. BBBC014 uses its 20 test FOVs; each external dataset is split into five validation fields for TV tuning and 15 held-out fields for evaluation. The model is trained only on BBBC014 channel 1 and external evaluation is zero-shot; TV deconvolution is tuned per dataset on its five validation fields.

Dataset	TV deconv	U-Net	Δ [95% CI]
BBBC014 (20 test FOVs)	35.34	36.70	+1.36 [+1.23, +1.49]
BBBC015 (15 fields)	34.63	35.15	+0.52 [+0.35, +0.68]
BBBC016 (15 fields)	33.81	35.10	+1.30 [+1.22, +1.37]
BBBC018 (15 fields)	25.62	25.78	+0.16 [+0.12, +0.21]

3.5. Real noisy/clean pairs

To test whether the restoration gain survives real noise, we trained the same U-Net topology (48 channels) on 1,951 real noisy/clean pairs from the AI4Life-MDC25 release and evaluated it on 243 held-out pairs from the same source (Fig. 4, Table 3). The network reaches 29.01 dB PSNR and 0.877 SSIM, compared with 24.83 dB and 0.559 for affine-calibrated tuned BM3D, 24.78 dB and 0.551 for NLM, and 16.44 dB for the noisy input. BM3D and NLM are within 0.05 dB of each other; the paired gain is 4.18 dB over BM3D (95% CI 3.94–4.42; exact Wilcoxon $p = 2.2e-41$), 4.23 dB over NLM, and 4.50 dB over TV denoising; the network wins on 240 of 243 pairs. A budget-matched DnCNN (16k patches, 80 epochs) reaches 28.80 dB and 0.865 SSIM, 3.97 dB above affine-calibrated BM3D and only 0.21 dB below the U-Net (95% CI 0.15–0.28; 210 of 243 pairs). The gain over classical methods is therefore shared by a second deep architecture, while the U-Net retains a small edge. This is an in-source real-noise result: training and test pairs come from the same acquisition protocol, with disjoint fields and no near-duplicate images. The real pairs contain no blur component, so they validate denoising within a matched protocol, not deblurring.

The FMD release was evaluated on a pre-specified stratified subset of 150 of its 301 held-out fields (SI S7b). With per-microscope-tuned baselines, the U-Net reaches 31.74 dB and 0.845 SSIM versus 28.07 dB and 0.704 for BM3D (+3.67 dB, 147/150; SI S7b). These numbers are not affine-calibrated, so FMD remains a secondary upper-bound check and is not used as a headline; the main real-data evidence is the nuclei set above.

When the same trained model is applied to the 20 large BBBC006 fields in the same release, the ranking reverses: the network reaches 15.83 dB, below the raw classical baselines BM3D (17.05 dB), NLM (16.89 dB), and TV (16.45 dB).

The large fields have 16-bit saturated references and a different noise model, so this is a cross-source transfer failure rather than a contradiction of the in-source result. It confirms that matching the training and deployment acquisition remains necessary, and it is the main reason we describe the dose sweep as a controlled simulation rather than a real low-dose validation.

Table 3: Real noisy/clean pairs from AI4Life-MDC25 (Jacquemet source). 243 held-out nuclei fields; classical parameters are tuned and affine-calibrated on validation fields; DnCNN is a second supervised deep baseline. PSNR in dB.

Method	PSNR (dB)	SSIM
AI4Life nuclei, 243 held-out fields		
Noisy input	16.44	0.107
TV denoising	24.51	0.526
NLM	24.78	0.551
BM3D	24.83	0.559
DnCNN (supervised)	28.80	0.865
U-Net	29.01	0.877

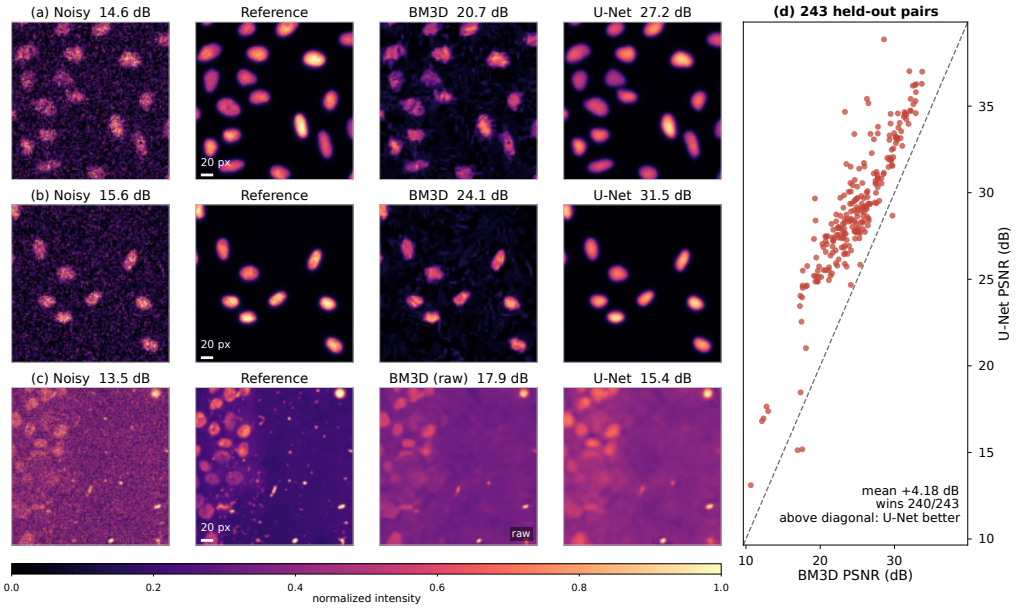


Figure 4: Real-data validation on held-out AI4Life-MDC25 pairs. (a,b) Two fixed held-out pairs at the 25th and 75th percentile of U-Net PSNR (median-quality examples, not cherry-picked best cases), showing the noisy input, clean reference, affine-calibrated BM3D, and U-Net; numbers are per-pair PSNR. (c) A large BBBC006 field from the same release (fixed field and crop), where the same weights fail under cross-source transfer; BM3D is raw here. (d) Per-pair PSNR of the U-Net versus BM3D for 243 held-out pairs; the dashed line is $y = x$, and 240 of 243 pairs lie above it. All image panels use one shared normalized intensity scale $[0, 1]$, shown by the single colorbar; scale bars are 20 pixels. The BM3D panel in (c) is raw and uncalibrated, as tagged in the panel.

3.6. Controls, PSF robustness, and speed

A phase-scrambled control with the same 64-channel/48k/150 budget, trained on surrogates with the same power spectrum but no spatial structure, reaches 30.19, 34.46, and 36.18 dB at $P = 3, 30, 100$, below both the real model (32.27, 36.70, and 38.61 dB) and TV deconvolution (31.34, 35.34, and 37.16 dB; Supplementary Information). The gain over the classical baseline therefore requires learned sample structure, not just extra denoising capacity.

The mixed-dose model tolerates a modest Gaussian-width mismatch: at $P = 30$ it reaches 36.92, 36.70, and 35.76 dB for $\sigma = 1.5, 2.0, 2.5$, compared with 35.85, 35.34, and 34.76 dB for TV deconvolution with λ retuned at each width (0.04 selected in all three cases; Supplementary Information); the network is ahead by

1.07, 1.36, and 1.00 dB at the three widths. The Gaussian-trained model does not transfer to a PSF with a different shape and width: with an in-focus Airy PSF (NA 1.4, 520 nm, 100 nm pixels; FWHM about 1.9 pixels, narrower than the $\sigma = 2$ training kernel), it reaches 32.63, 36.64, and 37.05 dB at $P = 3/30/300$, i.e. +1.42, +0.41, and -3.21 dB versus TV deconvolution with the correct PSF. A model trained on a 50/50 Gaussian/Airy mixture recovers the high-dose performance: 32.71, 37.61, and 41.55 dB (gains 1.50, 1.39, and 1.29 dB; 95% CIs 1.35–1.65, 1.28–1.48, and 1.13–1.45; 20/20 FOVs), while retaining 32.26, 36.60, and 39.83 dB on the Gaussian benchmark. The dose-agnostic result is therefore not tied to the exact Gaussian kernel, provided the training distribution covers the deployed PSF. An exploratory dose+PSF-conditioned model (32 channels, 20k patches, 60 epochs; it receives both the dose P and σ) reached 36.69, 36.03, and 35.44 dB at the same σ values, 0.23–0.67 dB below the strong blind model; because it is neither budget-matched nor blind, we do not adopt it (`results/dscond_eval.json`). Adding Gaussian read noise of 0.005 or 0.01 (1.3 or 2.6 8-bit counts) at $P = 30$ changed the network and fair TV values by at most 0.04 dB (SI).

Inference takes 1.2 ms per 128×128 patch on an A100 and 29.3 ms on one CPU core, versus 103 ms for 30-iteration ADMM TV deconvolution on one CPU core. The network is therefore 3.5× faster on the same CPU core and 86× faster when the network runs on the GPU. A full 1360×1024 FOV requires about 85 patches, i.e., about 0.1 s on the GPU or 2.5 s on one CPU core: this supports interactive and many online workflows, but not full-FOV frame-rate feedback on a single CPU core.

3.7. Adaptive allocation is target-dependent

We also tested whether a cheap preview and per-FOV uncertainty could choose a different dose for each field. An oracle with access to each FOV's full dose-response would save 42–71% of photons across the tested targets, but a ground-truth-free policy calibrated on validation FOVs did not consistently beat the best fixed dose at matched test success; details are in the Supplementary Information. We therefore recommend the fixed protocol dose read from Fig. 2e rather than per-FOV allocation.

4. Discussion

The gain has two sources. Denoising capacity accounts for part of it, but the phase-scrambled control shows that capacity alone does not beat fair TV deconvolution. The rest comes from a learned prior over sample structure: at

$P = 3$ the input retains little more than the position and approximate shape of a cell, yet the network recovers boundaries and internal texture that TV deconvolution smooths away. The same prior raises the achievable quality at high dose, where the classical method saturates. The dose-conditioned and specialist comparisons show that this prior can be learned from mixed doses without a dose input; the photon count is not needed at inference. Consistent with this, a linear probe on simple image statistics identifies the five dose levels with 97.9% accuracy (100% with a small CNN; SI S8), so the information needed to adapt the effective regularization is present in the noisy input.

With validation-selected doses the protocol yields a 3–3.3× lower photon budget at four targets and ties at three; two validation selections miss the held-out criterion, and the test-oracle upper bound is five savings and four ties. At SSIM 0.85 the classical method is already at the same dose, so there the network's advantage is the quality ceiling rather than the threshold. The cross-dataset gains suggest that the prior is not tied to one cell type or stain, although they are smaller outside the training distribution.

Simulation.. The noise is simulated as Gaussian-blurred Poisson noise with a known, spatially invariant PSF; camera read noise, background structure, and optical aberrations are absent. The high-SNR reference is itself diffraction-limited, so the network removes an additional synthetic Gaussian blur rather than the microscope PSF; the experiment tests photon efficiency under controlled blur, not a full optical reconstruction of an unknown object. The PSF tests cover Gaussian widths of 1.5–2.5 pixels and an in-focus Airy PSF; a PSF-mixed model retains the gain under Airy, but arbitrary, aberrated, or depth-varying PSFs remain untested. The simulation model is trained on one dataset and evaluated zero-shot on three others; the external gains are smaller, and training on a mixture may improve them.

Real data.. A downstream BBBC039 nuclei-mask check with per-dose tuned TV found only a small Dice gain (0.795 versus 0.773 at $P = 1$; SI), so the restoration advantage does not automatically translate into a large segmentation gain. The real-data check is in-source: on 243 held-out AI4Life pairs the real-data model beats tuned BM3D by 4.18 dB, but the same weights fail on the large BBBC006 fields in the same release, which differ in bit depth, saturation, and noise model. Real paired data therefore support restoration within a matched acquisition protocol, not the cross-source dose claim. A label-free J-invariant variant [33] trained only on the noisy images reached 17.89 dB on the held-out set, below tuned BM3D (21.30 dB, uncalibrated), so the paired clean references are necessary for the present gain.

The FMD check uses a pre-specified 150-field subset and requires retraining on FMD; it is not a zero-shot transfer, and the remaining 151 test fields were not evaluated.

Uncertainty and hallucination.. The uncertainty ensemble uses the conditioned variant; because conditioning changes accuracy only marginally, we expect the calibration to transfer, but this is not tested directly. Learned restoration can produce plausible but incorrect structure [34]; uncertainty flags unreliable fields but does not prove that a confident reconstruction is artifact-free. Validation on matched real low-dose data remains necessary before quantitative biological use.

The closest prior work differs in mechanism and evidence. FFDNet conditions a denoising network on a noise level, and CARE learns a supervised restoration mapping from paired images. Here a single mixed-dose model needs neither a noise-level input nor per-dose retraining, and it is within 0.05 dB of the conditioned variant. Ye et al. [19] use uncertainty to drive adaptive rescanning; our protocol is a fixed widefield acquisition, and an adaptive extension is reported as a negative result in the SI. Liu et al. [17] use a model-based deep prior, whereas our network is trained end-to-end across dose. Qiao et al. [18] remove the need for paired clean targets by zero-shot learning from noisy images; that addresses the supervision axis, whereas the matched-success protocol here addresses the dose axis and measures the photon-budget gain against a per-dose-tuned classical baseline. Sheneman et al. [20] demonstrate a hardware near-zero-photon microscope, whereas this is a controlled simulation of the dose sweep with an in-source real-data check.

Operationally, a laboratory would train once over the dose range of interest, read a fixed dose from the protocol curve for the desired success rate, and use ensemble uncertainty to flag fields that need more light. The approach requires no change to the microscope and no photon-count metadata at inference; the simulation model is trained once across the dose range, and the real-data model is trained once on paired images from the deployment protocol. Its speed is compatible with online processing budgets.

Conclusions

One mixed-dose U-Net, with no photon-count input, matches or beats a fully tuned TV-ADMM baseline across a 100-fold dose range and, at matched success, uses a threefold lower photon budget. On real paired data it improves over affine-calibrated BM3D by more than 4 dB and wins on all but three fields. The

gain is not a capacity effect: conditioning on dose adds less than 0.05 dB, and a budget-matched phase-scrambled control remains below TV. For a fixed acquisition protocol, these results argue that dose metadata and per-dose retuning are not required, and that restoration methods should be compared by the photon budget needed at a matched success rate. The next step is prospective validation on real low-dose acquisitions and other microscope PSFs.

CRediT authorship contribution statement

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Data and code availability

All datasets are public: BBBC014, BBBC015, BBBC016, BBBC018, and BBBC039 from the Broad Bioimage Benchmark Collection [21], AI4Life-MDC25-Nuclei [22] (Zenodo 15624741, CC BY 4.0), and AI4Life-MDC25-FMD [24] (Zenodo 15761900, CC BY 4.0), with accession numbers and licenses listed in the Methods. Code for data splits, training, baselines, statistics, and figures, together with the JSON result files and checkpoints, will be released with the manuscript.

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